BUILT SO FAR

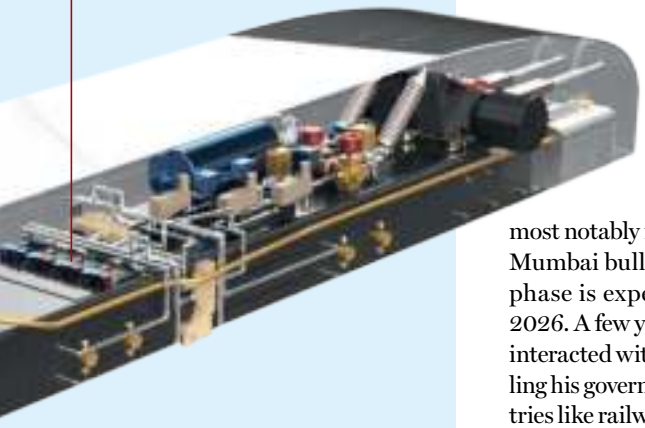
422-metre-long, 2-metre-diameter vacuum tube: The world's largest student-built Hyperloop testbed

Pod prototypes developed: Currently progressing to Version 8.0

Onboard systems under testing: Life support, humidity control, thermal regulation

Concrete tube prototype in progress: Projected 80% cost reduction

MISSION CONTROL GUI
Real-time diagnostics, remote monitoring



THE CHALLENGES

- No full-scale working Hyperloop demonstrator yet
- Concrete tube still in R&D phase; vacuum retention is a key hurdle
- Passenger applications likely several years away; safety validation a big question mark
- Global commercial viability likely post-2040, say some analysts

are still unsolved: maintaining vacuum in tubes, controlling levitating pods at high speeds and ensuring passenger safety. And, as with most infrastructure projects in India, land acquisition for long-distance corridors could become a stumbling block. With no established safety standards, regulatory approval will be another hurdle. Officials say they'll cross that bridge when they come to it. "Let's applaud the ambition. Let Indian Railways experiment boldly," says Sudhanshu Mani, former GM, Integral Coach Factory (ICF) Chennai, "but let's not confuse the hopeful 400-metre scaled-down prototypes with revolutions."

THE ROAD AHEAD

High-speed transport is a part of the Narendra Modi government's infrastructure playbook. Their fascination with cutting-edge mobility solutions is well-known—most notably in the ongoing Ahmedabad-Mumbai bullet train project, whose first phase is expected to launch by August 2026. A few years ago, the prime minister interacted with the Avishkar team, signalling his government's serious intent. Ministries like railways and even ports, shipping and waterways are in active consultations to support proof-of-concept use-cases for cargo. Government entities have indicated that the full-fledged rollout should not cost more than metro systems, approximately Rs 200 crore per km. The cargo-movement solutions, however, will cost much less than that, officials say. "The entire tech that is used is indigenous, which is a matter of pride. In the railway set-up, we will decide on a site for commercial transportation over a distance of 40-50 km. ICF Chennai will make its electricals," says Vaishnaw.

Indian Railways has already signed a pact with IIT Madras to set up a Centre of Excellence for Hyperloop technology. The objective is to develop a sub-scale model of the pod, along with a test track and vacuum tube facility at the institute. Chennai-based startup Tutr Hyperloop, in which IIT Madras has a stake, is hastening the commercialisation of the technology. TuTr Hyperloop, Technical University of Munich (TUM) and Neoways Technologies GmbH have signed an MoU to drive the research, development and deployment of Hyperloop technology.

Says Prof. Satya Chakravarthy, advisor (Hyperloop), IIT Madras: "By partnering with TuTr Hyperloop, TUM and Neoways, we are moving closer to making Hyperloop a reality." TuTr has also partnered with global transport engineering firm SYSTRA, with plans to jointly launch a pilot project. "If we can transport cargo even at 150-200 kmph, that's over three times the average speed of trucks. That makes it a business case in India," says Aravind Bharadwaj, CEO, TuTr.

Government entities say a full rollout should cost no more than metro systems—around Rs 200 crore per km—while cargo solutions will be significantly cheaper

Next up: the long test track, an improved pod with cargo that will move an item of 100 kg by the year-end. India Inc has lent a hand as well. Tutr calls it a consortium approach, collaborating with firms such as L&T Construction, ArcelorMittal and Tube Investments of India. "Every one in the consortium is an expert in their field. Tutr is bringing all that together," Bharadwaj says.

Official sources say talks are on to trial parcel/ cargo movement in railways, metro and ports, even if on a small scale, starting sometime this year. "We can alleviate port congestion by hauling cargo," Prem claims. "Our ultimate goal is that the first Avishkar Hyperloop will be a 350-km tube between Chennai and Bengaluru, making the travel possible in 30 minutes."

No one can predict how far the Hyperloop dream will travel in times to come. But this spirited band of young innovators, coupled with an agile startup, has already ensured that India won't be left behind for lack of ambition or effort. ■

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